Lesson plan · for the teacher

Models with numbers: unit check

Optimization · Level 2 · Unit check

Time	Materials
40 min	Board, chalk, paper. No devices needed.

Learning goals

- From words to a model
- The best plan is at a corner
- Whole numbers and yes-or-no choices
- Your first model in Python

Activity: Team quiz

1. Teams (3 min): Make teams of four and give each a sheet for its answers.
2. Quiz (15 min): Read each question aloud; teams talk it through and write one answer.
3. Explain (7 min): Reveal each answer and ask a different team to explain why it is right, using the words variable, constraint and objective.
4. Model it (10 min): Each team picks a school decision (duties, a purchase, the garden) and writes its variables, constraints and objective on paper.
5. Wrap-up (5 min): Write the question most teams missed on the board, and start tomorrow's lesson with it.

Discussion

- Which idea from this unit will you use first in class?
- Where could rounding mislead us at school?
- Who should help write the rules of a school model?

Exit ticket

- One decision I will model with my class, and its objective.

Try it in class tomorrow
Pick one decision at your school and write it as a model with a colleague: the variables with units, the constraints and the objective. Check one plan by putting in its numbers, and ask who the model leaves out.

Worksheet · for pupils

Models with numbers: unit check

Optimization · Level 2 · Unit check

Name _____ Class _____ Date _____

1 **In the school lunch model, which part is the objective?**

- ◯ Cups of rice per pupil
- ◯ Iron at least 5 mg
- ◯ The total cost, to make as small as possible
- ◯ The price of a cup of rice

2 **Match each piece of the lunch model to what it means.**

energy ≥ 700	• • • •	A variable: cups per pupil
nuts ≤ 2	• • • •	The cost, to minimise
cost = 15 × rice + … + 8 × nuts	• • • •	At most 2 handfuls of groundnuts
rice	• • • •	Enough energy

3 **The garden plan 6 tomato beds and 2 bean beds keeps all three limits (x + y ≤ 8, 2x + y ≤ 14, x + 3y ≤ 18).**

- ◯ True
- ◯ False

4 **Why does the simplex method only visit corners of the feasible region?**

- ◯ Plans inside the region break a limit
- ◯ The best plan of a linear program is always at a corner
- ◯ Corners are easier to draw

5 **Which decisions need whole-number or yes-or-no variables? Choose all that apply.**

- ☐ How many buses to hire
- ☐ Whether to fund the chess club
- ☐ Litres of paint for the classroom
- ☐ How many teachers to hire
- ☐ Kilograms of rice for the kitchen